XFMC Quick Tutorial

Version 3.52 – December 2025 Example Flight: KPDX → KSJC

Requirements

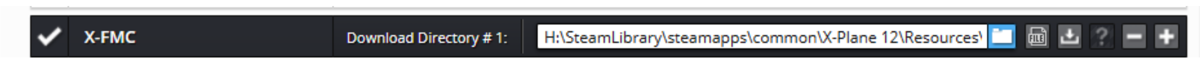
1. **X-Plane 12** (working installation)
2. **Aircraft**: x737 by Benedikt Stratmann □ Download here (or standard Laminar 737)
3. *(Optional)* **Better Pushback plugin** □ Download here
4. *(Optional but recommended)* **Little Navmap** □ Download here
5. **XFMC fully installed** □ GitHub repository

Setting Up Navdata

6. **Navigraph Navdata (SID/STAR)**
   - Requires a subscription (one-time payment possible).
   - Navigraph downloads
   - Select **XFMC** format from Navigraph downloads.
   - Extract the provided *xfmc native zip* into the folder: `XFMC/Navdata`.
   - x-fmc_native_2413.zip 22.769.355 19.589.209 WinRAR ZIP-archief 6-1-2025 15:25 2F73AEAA
   - After unzipping, your `xfmc/navdata` folder should contain the updated files.

Sids	10-5-2025 21:34	Bestandsmap	
Stars	6-7-2025 19:16	Bestandsmap	
airports.txt	17-12-2024 08:54	Tekstdocument	2.735 kB
ats.txt	17-12-2024 08:54	Tekstdocument	11.042 kB
cycle.json	17-12-2024 15:41	JSON File	1 kB
cycle_info.txt	17-12-2024 10:16	Tekstdocument	1 kB
navaids.txt	17-12-2024 08:54	Tekstdocument	697 kB
waypoints.txt	17-12-2024 08:54	Tekstdocument	8.444 kB

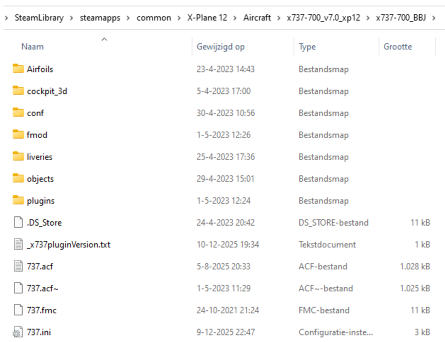
7. **SimBrief Flightplan**
   - Register a free account at SimBrief.
   - Generate and download a flightplan in XFMC format.
   - In the simbriefdownloader select XFMC and save it to: `XFMC/Flightplans`.



Creating the Flightplan

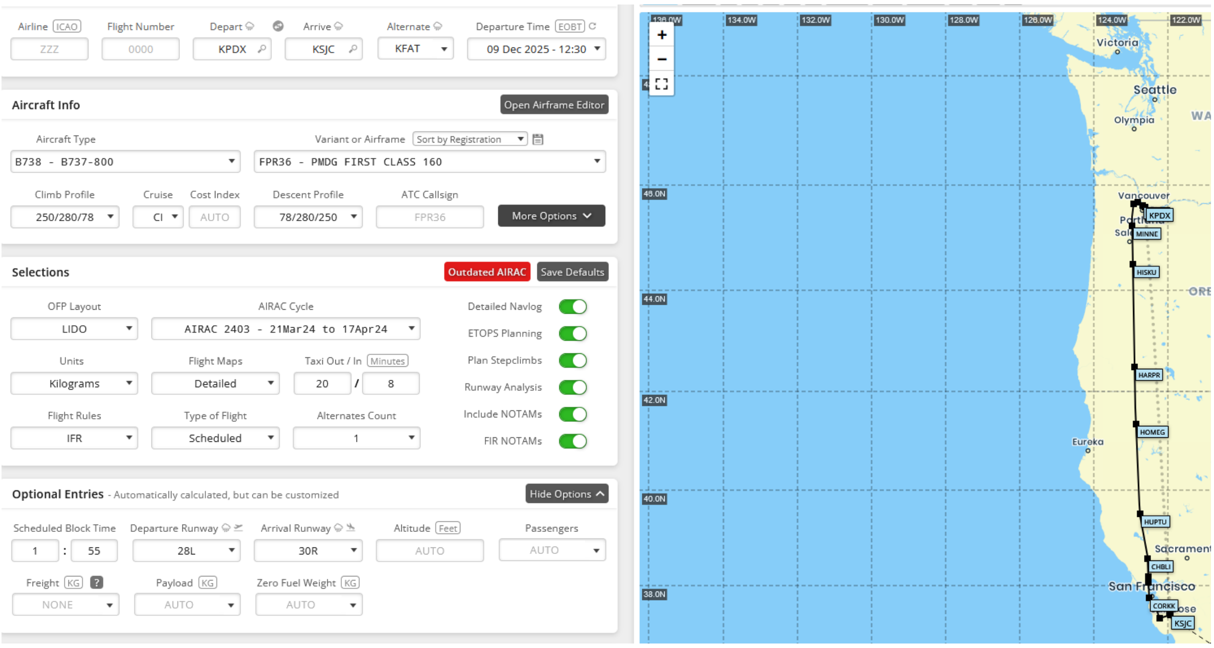
- Departure: **KPDX (Portland)**, Runway **28L**
- Destination: **KSJC (San Jose)**, Runway **30L**

Aircraft: **Boeing 737**

-  **You need the 737.ini config file for XFMC, which should be installed in the x737 directory** 

Click **GENERATE** in SimBrief. The route will be saved to …`/XFMC/Flightplan` as:

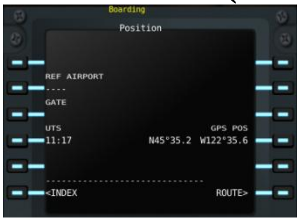
MINNE5 HISKU Q5 HUPTU CHBLI BRIXX4



XFMC Setup

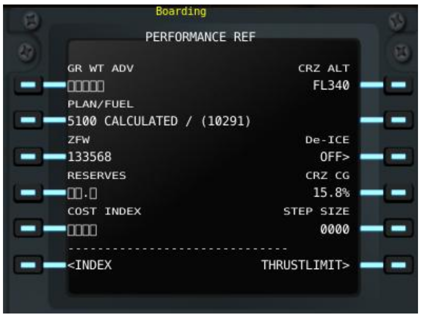
1. Select **FMC**. 

2. Go to **POS INIT (RSK6)** → then **ROUTE (RSK6)**.

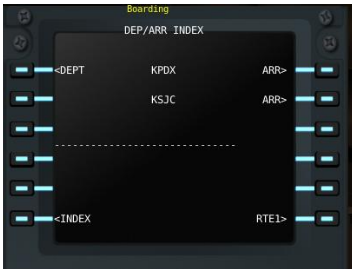


   - Enter `KPDXKSJC` in the console.

   - Click **CO ROUTE**, then **EXE**.



3. Open the **Performance Page**.
   - Enter flight level: `340` → **EXE**.



4. Press **DEPARR** → **LSK1 (DEPT)**.

5. Select **Runway 28L** → **EXE**.

6. Choose **SID MINNE5**/HITSKU (use **NXT PAGE**) → **EXE**.

7. Insert arrival details early (for TOD calculation).
    - Destination: **KSJC**, Runway **30L**.

Select STAR **BRIXX4** transition arrival: **JILNA**, choose **ILS**, then **EXE**. **(Study the arrival STAR KSJC see appendix)**

8. Verify the **Leg List**.
    - Scroll through with up/down keys.
    -  Ensure no `[DISCON]` messages (indicates discontinuity in route).
    -

9. In the x737, set altitude dial to **34,000 ft**.

    - Shortcut: type `340` in XFMC command line → click **MCP**.

10. Go to **INIT → TAKEOFF**.
    - Enter flaps: `1` → **LSK1**.
    - Set V-speeds: **V1 (RSK1)**, **V2 (RSK3)**, **Vref (RSK2)**.
11. Set flaps to **first detent**.
12. Release parking brake and taxi to **Runway 28L (KPDX)**.

□ You are now ready for departure!

Takeoff and Climb

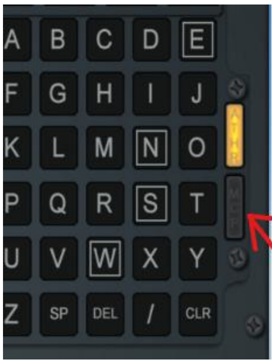
1. **At the holding point**
    - Click on **AP XFMC** and enter **Runway 28L**.
    - Align the aircraft to the centerline.
    - Perform a final check — everything should be OK.
    - Apply **full throttle**, keep the aircraft on the centerline.
    - When XFMC displays or calls **"PULL UP"**, gently pull the joystick.
    - The aircraft will lift off. Continue climbing until you see the message **"INIT CLIMB"**.
    - Gently release the joystick — XFMC will take over.
2. **Neutral joystick position**
    - Once the joystick is neutral, you are done for now.
    - XFMC will automatically follow the **SID** and continue en route to MINNES.

3. **Top of Climb (TOC)**
   - After around 70nm ,just after passing HISKU the aircraft will stop climbing at **34,000 ft**.
   - This marks the **TOC**.
   - The aircraft will then enter **cruise mode**.
   -
   - TOC can be observed in VNAV climb page ….

   - Your x737 panel should look like this during climb .Note the AP1 and AP2 lights are **OFF**, and need to be OFF. **XFMC is running is dedicated AP mode!**

Cruise

- If you want to change the flight level:
  - Type the new FL in the XFMC command line (e.g., `360`).

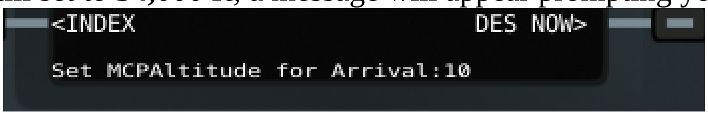


- Click on **MCP**.
- XFMC will then climb or descend to the new flight level.
-
- If you want to change the cruisespeed: Vnavpage Cruise: change econ speed

Descent Phase

Top of Descent (TOD)

- When the **Top of Descent (TOD)** is reached, the aircraft will begin its descent.
- If the altitude is still set to **34,000 ft**, a message will appear prompting you to reset the



MCP for arrival.
- Set the MCP to **2000 ft or lower**.
- At this point, your x737 panel should look like this when the descent starts.

Early Descent

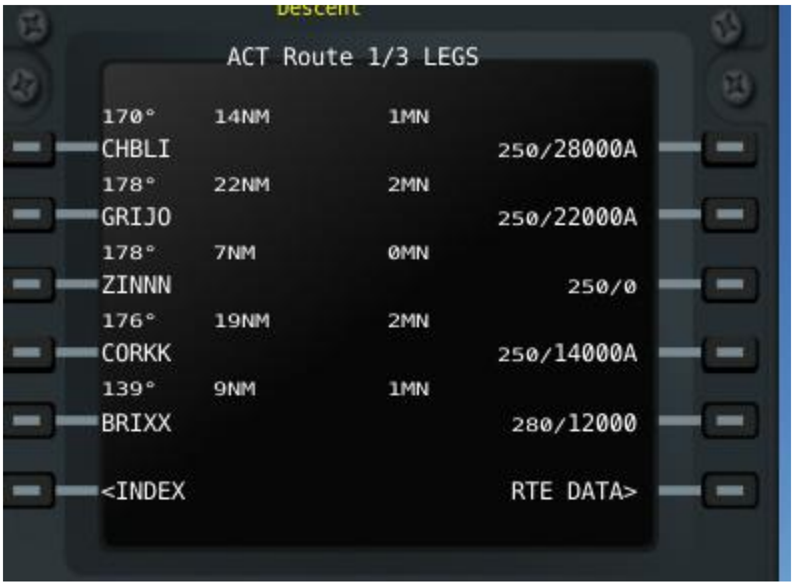
- An early descent is possible within **60 NM** of TOD.

- To initiate, go to **VNAV page 3** and click **DESCENT NOW**.
- 

- **Descent Inhibit Logic** At certain points, **XFMC** may temporarily halt the descent.
  - This is due to **Decision Inhibit logic**, which delays descent to prevent excessively shallow profiles.
- **ATC Instructions**
  - Normally, when **CTR** is active, the pilot reports: *"FPR ready to descend."*
  - If CTR issues instructions for a specific flight level, simply enter the flight level in the command line and click **MCP**.
- **Speed Management**

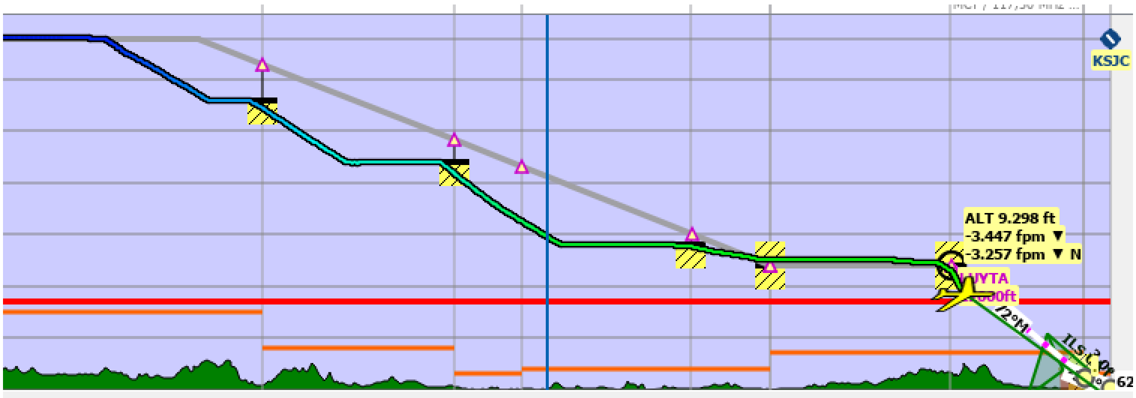
XFMC Leg List During Descent

- At the first waypoint **CHBLI**, XFMC will level off at **28,000 ft**.
- Because the descent is stepped, it is recommended to reduce your speed to **260 kts**.
  - Go to **VNAV page 3** and change the **ECON Speed** to 260.
- At the STAR point **CORKK**, XFMC will level off around **14,000 ft**.
- During the STAR descent, the profile is also **stepped**: each waypoint must be crossed at a specific altitude.

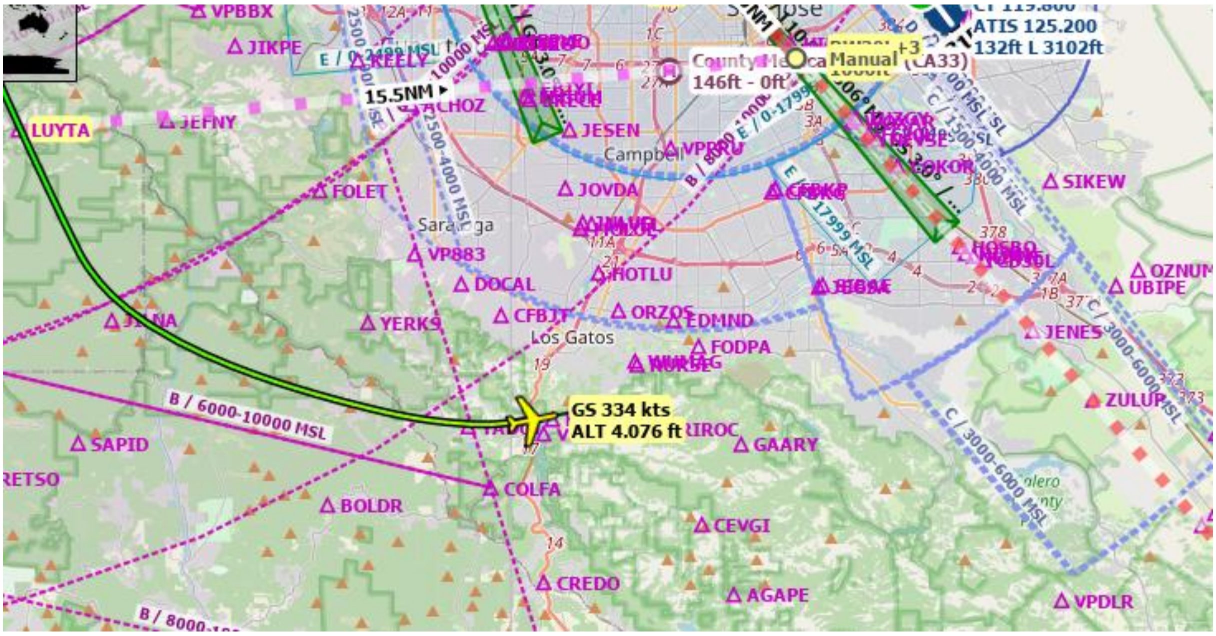


At STAR point CORKK XFMC Will level off around 14000 feet

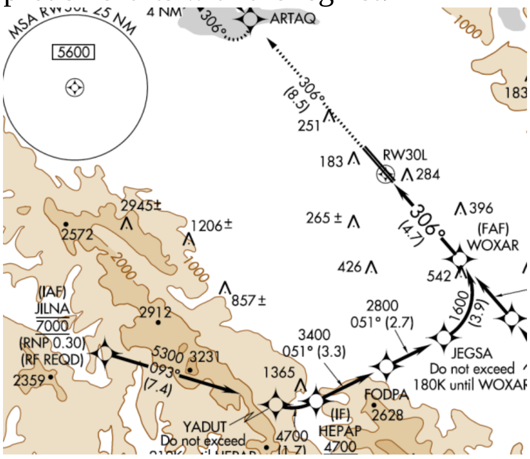
During the descent the star is stepped, each waypoint should be reached at a certain altitude:



Approach Procedure Verification



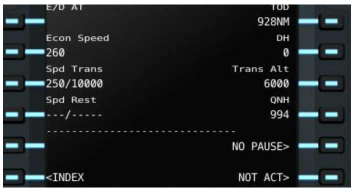
1. **Verify the procedure** Compare the approach charts with the leg list.

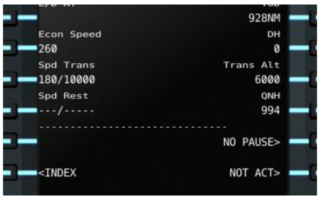


2.

- All points match the chart, so the arrival is safe.
- Since the approach is between mountains, extra caution is required.
- Activate the **APP** button as soon as the aircraft turns inbound at **WOXAR**.
- This is a **short arrival**, so reduce speed at **YADUT** (or before).
- On the **VNAV page**, change the transition speed from *250/10000* to *180/* and confirm by clicking on the line.

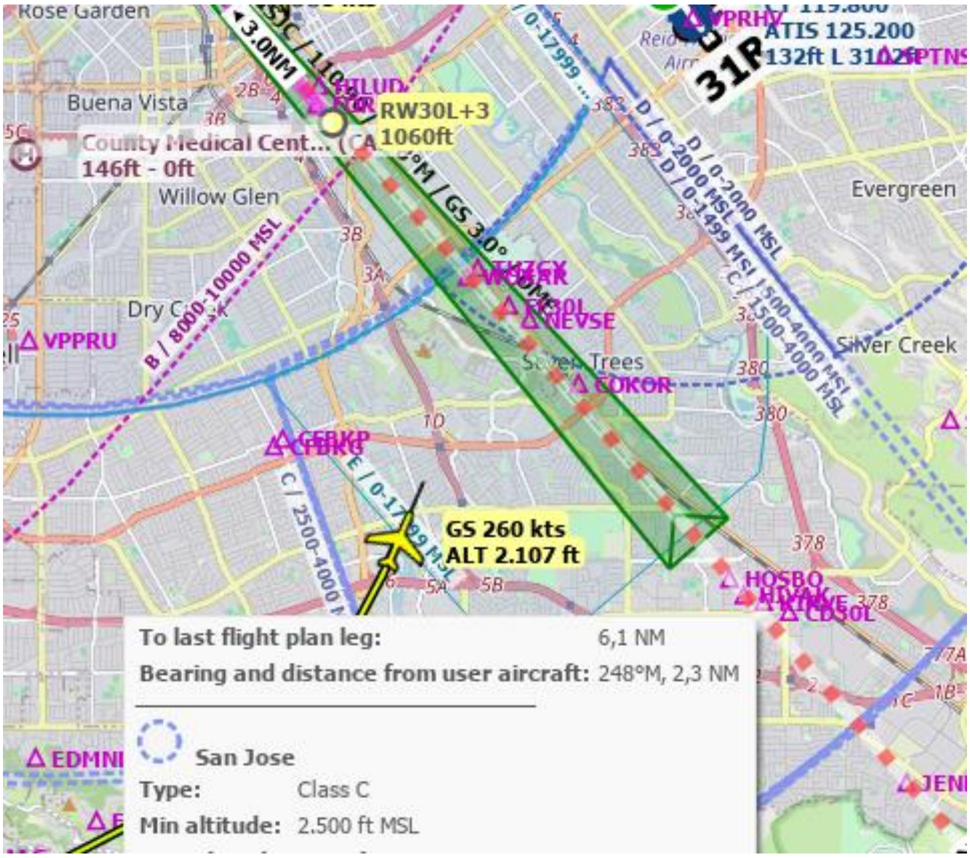
3. Select the ILS frequency before entering the cone: (click on LSK4)

- 

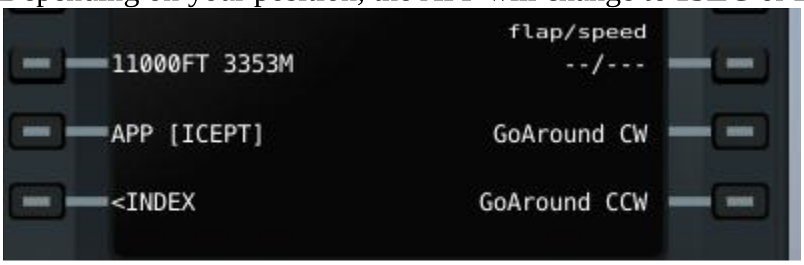


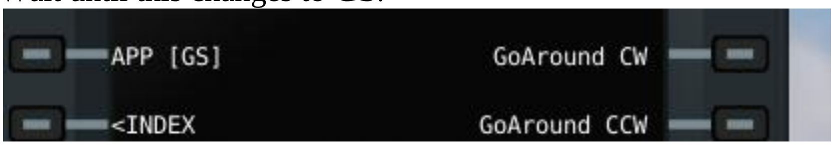
4.  **Manual speed control (alternative, RECOMMENDED)**
   - Control speed manually using the throttle.
   - Disable **AHTR** on **XFMC**.
5. **Aircraft speed control (alternative)**
   - Disable **AHTR** on **XFMC**.
   - Switch **A/T ARM ON , click SPD button ON**

---→ At WOXAR reduce speed to 160 knots.

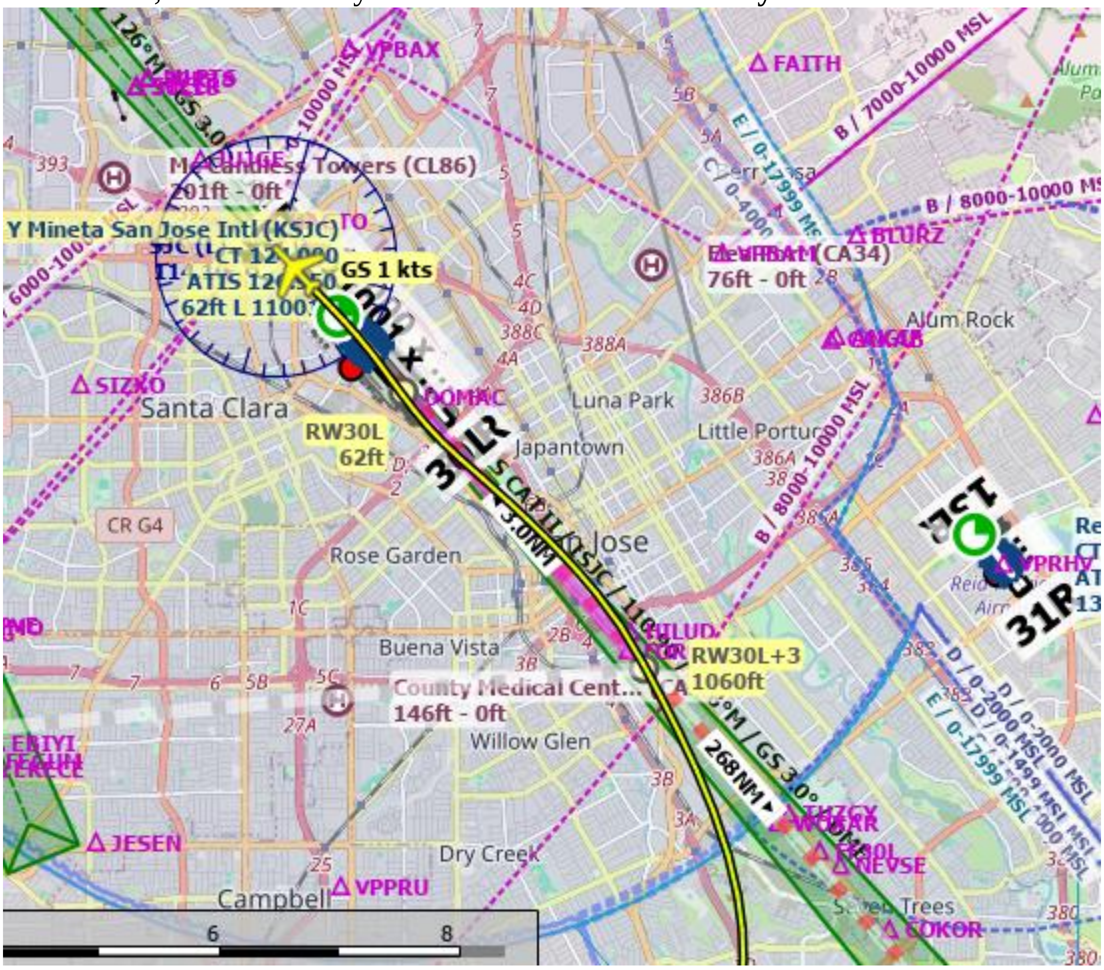


Arrival

1. **Arrival Page**
   - Go to the **Arrival Page**.
   - Click on the **APP** line.
   - Depending on your position, the APP will change to **ISEC** or **LLZ**.
   - 

2. **LOC Beam**
   - If the aircraft is on the **LOC beam**, **LLZ** will be displayed.
   - Wait until this changes to **GS**.
   - 

3. **Glide Slope Capture**
   - Once **GS** is active, the aircraft will begin descending.
   - Set the **first flap detent**.
   - Reduce speed to **160 knots** using manual throttle, the x737, or via the VNAV page 3(transition speed/altitude: type `160/`).

4. **Flaps Deployment**
   - Continue deploying flaps step by step until reaching **maximum flaps**.
5. **Landing Gear**
   - At **800 ft AGL**, deploy the landing gear.
6. **Decision Height (DH)**
   - At **200 ft AGL**, the **DH point** is reached.
   - This is the last point to initiate a **go-around** if necessary.
7. **Flare**
   - At **30 ft AGL**, the aircraft will flare.
   - Nose pitches up slightly, and the wheels make contact with the runway.
8. **Braking**
   - Engage **reverse thrust** and apply brakes.
   - Keep the nose aligned with the centerline.
9. **Taxi Off Runway**
   - At **20 knots**, exit the runway via the **first available taxiway**.
   - 

 You have now completed the arrival and landing sequence.!

“This arrival is considered one of the most difficult.”

Situation: When ATC Pops Up

1. **Initial Vectors** ATC will provide vectors:
   - *[FPR36]* “Turn left heading 090, descend to 8000 feet. Expect runway 12R. Reduce speed to 240.”
2. **Heading Control**
   - Disable **LNAV** on **XFMC**.
   - On the **x737**, set the heading to the assigned value.
   - Alternatively, type `HEADING/xxx` in the XFMC command line and click **FIX**.
3. **Runway and Frequency Setup**
   - On the **ARR page**, select runway **12R**.
   - Go to the **Radio page** and set the ILS frequency.
4. **Speed Management**
   - On **VNAV page 3**, enter the transition speed: `240/`.
   - Alternatively, disable **ATHR** on XFMC and set speed manually on the x737.
5. **Additional Vectors**
   - When ATC provides further vectors, follow the same procedure as above.
6. **ILS Clearance**
   - When ATC instructs: *[FPR36]* “Turn right heading 120, you are cleared for ILS approach runway 12R KSJC. Report when established on the localizer.”
   - Go to the **Arrival page**, click **APP**.
   - **ICEPT** or **LLZ** will appear.
7. **Glideslope Capture**
   - As soon as **APP [GS]** is displayed, report to ATC: *[FPR36]*

This tutorial is AI-generated, reviewed and controlled by a human

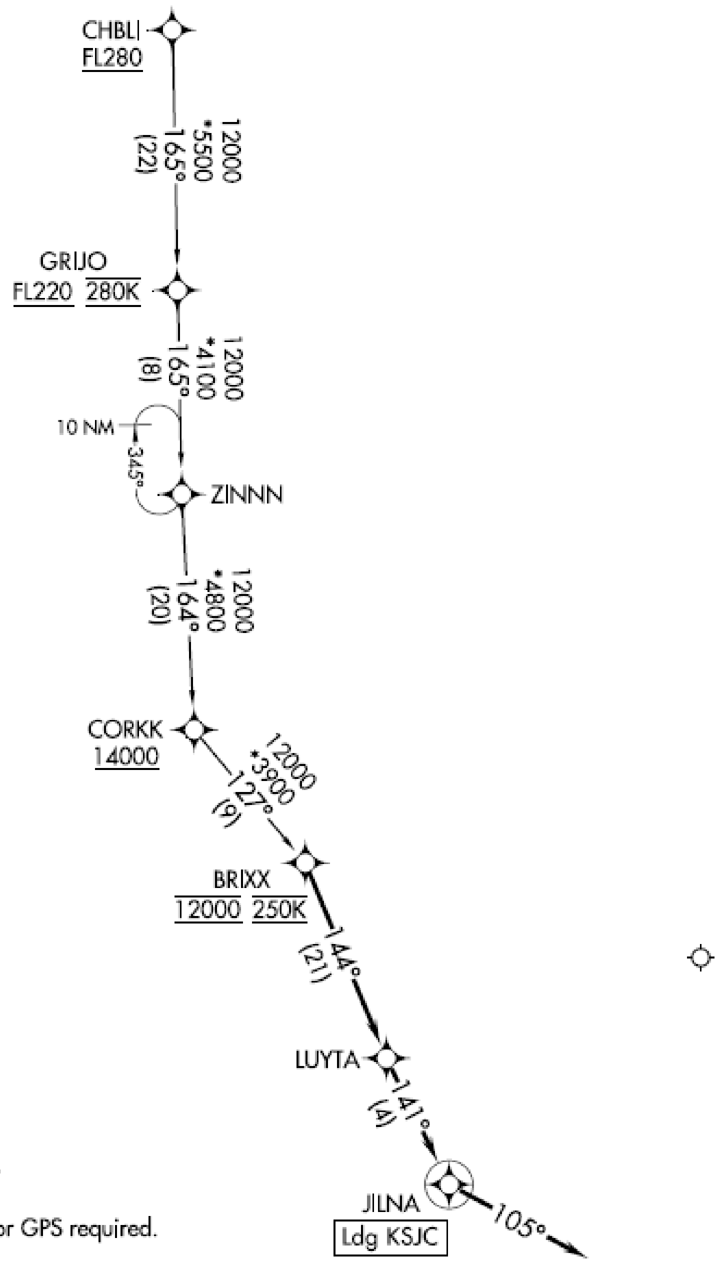
(BRIXX.BRIXX3) 21168

BRIXX THREE ARRIVAL (RNAV)

AL-693 (FAA)

NORMAN Y MINETA SAN JOSE INTL (SJC)
SAN JOSE, CALIFORNIA

OAKLAND CENTER
125.85 323.0
NORCAL APP CON
133.95 317.6
D-ATIS
126.95
GND CON
121.7
SAN JOSE TOWER★
124.0 257.6



NOTE: RADAR required.
NOTE: RNAV 1.
NOTE: DME/DME/IRU or GPS required.

NOTE: Chart not to scale.

ARRIVAL ROUTE DESCRIPTION

SW-2, 22 FEB 2024 to 21 MAR 2024

SW-2, 22 FEB 2024 to 21 MAR 2024